

Survey on the Integration of Distributed Learning and Transformers

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March 3, 2025

1 Background and Motivation

This week’s research focuses on the combination of federated learning, split learning and transformer architectures. As Transformer models demonstrate remarkable performance across various tasks, exploring how to effectively integrate them into the FL framework has become an increasingly important topic. The goal is to leverage the superior representation learning capability of Transformers while meeting the unique requirements of FL, such as privacy protection and decentralized training.

Most existing FL research focuses on traditional architectures such as CNNs and RNNs, while adaptations and optimizations for Transformers in FL settings are still in the early stages. The deployment of Transformers in FL faces several critical challenges, including heavy computational costs, high communication overhead, and degraded generalization performance under non-IID data distributions. These challenges form the core focus of this week’s literature review.

2 Recent Advances in Applying Transformers to Distributed Learning Systems

2.1 New Framework on Federated Learning

2.1.1 TransFed: A way to Epitomize focal modulation using transformer-based federated learning[1]

This paper aims to improve the performance of Focal Modulation based Transformer model in federated learning under the scenario of data heterogeneity (non-IID), and solve the negative impact of FedAvg algorithm on focus modulation. The challenge is the inadequate adaptability of focus modulation in a federated learning environment. In this paper, a TransFed framework is proposed to generate personalized focus modulation parameters for each client by introducing learning-to-Tailor and server-side hypernetworks to effectively deal with data heterogeneity and improve model performance.

2.1.2 Federated Learning with Dynamic Transformer for Text to Speech[2]

This paper aims to address the challenges of training Text-to-speech (TTS) models using federated learning. The main challenge is the slow and unstable convergence of transformer models in a federated learning setting. To tackle these issues, the authors propose a Federated Dynamic Transformer, which significantly improves performance compared to federated transformers and approaches the performance of centralized Transformer-TTS models as the number of clients increases. The method also achieves faster and more stable convergence while reducing communication time.

2.1.3 FedPrefix: Towards Partial Model Personalization of Vision Transformers in Federated Learning[3]

The goal of this paper is to explore how partial model personalization can be applied to ViT to enhance the efficiency of Personalized Federated Learning (PFL). While previous work on partial model personalization has primarily focused on Convolutional Neural Networks (CNNs), there has been

limited research on its application to ViTs. Through empirical analysis of the sensitivity of different layers to data distribution, the authors identify that the self-attention layer and classification head are the most sensitive. Based on this insight, they propose a novel approach called FedPerfix, which uses plugins to transfer information from the aggregated model to local clients for personalization.

2.1.4 FedYolo: Augmenting Federated Learning with Pretrained Transformers[4]

This paper aims to address the challenges of learning with mobile and edge devices in federated learning by leveraging pretrained transformer models (PTFs). The key goals are to enable clients to use a single general-purpose model for diverse tasks while managing device and network constraints. The challenges include heterogeneity in client goals, and communication bottlenecks. The authors propose a method that focuses on two factors: model scale and modularity. A larger model scale reduces accuracy gaps and improves robustness, allowing for more local learning and fewer communication rounds. Modularity reduces communication overhead by over 100 times, boosts local adaptation generalization, and prevents catastrophic forgetting. The solution, "You Only Load Once" (FedYolo), enables efficient learning by loading a single PTF model and using modular updates for each task, ensuring minimal forgetting and high communication efficiency.

2.1.5 Training Vision Transformers in Federated Learning with Limited Edge-Device Resources[5]

The article's goal is to address the challenge of applying vision transformers (ViTs) in federated learning on resource-constrained edge devices. It introduces FedVKD, a federated knowledge distillation training algorithm that allows small convolutional neural networks on edge nodes to periodically transfer knowledge to a large server-side transformer encoder. This method reduces the computational load on edge devices while leveraging the powerful ViT structure. The FedVKD algorithm outperforms FedGKT and FedAvg, particularly on larger datasets with non-IID variations.

2.2 Theoretical Research on Federated Learning

2.2.1 FedTP: Federated Learning by Transformer Personalization[6]

This paper explores ways to maintain transformer model accuracy in Federated learning (FL) scenarios, mainly using ViT networks. ViT is similar to CNN and ResNet-18 in that it is used for image classification tasks. The main challenge this paper faced is that aggregation algorithms represented by FedAvg will destroy the self-attention mechanism of the transformer. Therefore, this paper proposed FedTP framework to apply transformer in federated learning. The self-attention layer in FedTP is generated by hypernetwork. Other layers in the model and hypernetworks are trained using FedAvg.

2.2.2 Rethinking Architecture Design for Tackling Data Heterogeneity in Federated Learning[7]

This paper aims to address the challenges of convergence and catastrophic forgetting in federated learning, particularly when dealing with heterogeneous devices and data. The authors demonstrate that self-attention-based architectures like Transformers are more resilient to distribution shifts, thus improving the performance of federated learning. They conduct a thorough empirical study comparing various neural architectures across federated algorithms and real-world benchmarks. The results show that replacing convolutional networks with Transformers significantly reduces catastrophic forgetting, accelerates convergence, and leads to a better global model, especially with heterogeneous data.

2.3 New Framework on Split Learning

2.3.1 Federated Split Vision Transformer for COVID-19 CXR Diagnosis using Task-Agnostic Training[8]

The paper aims to address the challenge of training neural networks in federated learning, particularly in healthcare, where data privacy is crucial, such as in COVID-19 diagnosis using chest X-ray images. The main challenge is the high network bandwidth consumption when exchanging weights in highly expressive architectures. The authors propose using split learning, which divides the network into

client and server parts to reduce computational load and bandwidth usage. They further enhance this approach by introducing ViT, which is well-suited for split learning due to its decomposable configuration, maintaining performance even with non-IID data.

2.3.2 FeSViBS: Federated Split Learning of Vision Transformer with Block Sampling[9]

The paper’s goal is to address the challenge of data scarcity in healthcare applications by proposing a framework called FeSViBS (Federated Split learning of Vision transformer with Block Sampling), which improves medical imaging classification tasks through federated and split learning. The proposed method introduces a block sampling module in the ViT, which samples intermediate features, distills them into pseudo class tokens, and sends them back to the client for effective feature augmentation. This approach enhances the model’s generalizability and outperforms other SL and FL methods on three medical imaging datasets (HAM1000, BloodMNIST, and Fed-ISIC2019) in both IID and non-IID settings.

2.4 Privacy on Split Learning

2.4.1 Visual Transformer Meets CutMix for Improved Accuracy, Communication Efficiency, and Data Privacy in Split Learning[10]

This article aims to propose a distributed learning solution for visual transformer (ViT) architectures, which face challenges in federated learning (FL) due to their large model sizes and computational demands. The issue is further compounded by the inefficiency of Split Learning (SL) when applied to ViTs, as the smashed data generated can be as large as the input data, reducing communication efficiency and threatening privacy. To address these problems, the article introduces a new method called "CutSmashed" data, which compresses and randomly punches the smashed data to improve efficiency and privacy.

2.4.2 Differentially Private CutMix for Split Learning with Vision Transformer[11]

The article aims to address the challenges of privacy and communication efficiency in federated learning when applying ViT due to their large model size and high computing costs. The authors propose DP-CutMixSL, a differentially private (DP) split learning framework that incorporates a novel privacy-preserving interpolation method called DP-CutMix. This method replaces randomly selected patches in smashed data to enhance privacy protection and reduce communication costs. The proposed approach not only improves privacy and communication efficiency but also achieves higher accuracy compared to Vanilla Split Learning.

2.4.3 Privacy-Preserving Split Learning via Patch Shuffling over Transformers[12]

The goal of this paper is to address the privacy-preserving issue in split learning. In traditional split learning, a neural network is split across different devices for training, risking the leakage of private training data. The authors propose an innovative patch shuffling scheme on transformers to preserve data privacy without compromising the overall model performance. Formal privacy guarantees are provided, and additional batch shuffling and spectral shuffling schemes are introduced to strengthen the privacy protection.

3 Conclusion

In my research, I focused on Vision Transformers (ViT). Several papers discussing the combination of federated learning and transformers highlight that the self-attention layer can be compromised by aggregation algorithms. In studies on the integration of partition learning and transformers, the issue of privacy is frequently addressed, suggesting that transformer segmentation is not the primary challenge. Furthermore, there are already several frameworks available for applying transformers in distributed learning environments.

References

- [1] T. Ashraf, F. Bin Afzal Mir, and I. A. Gillani, “Transfed: A way to epitomize focal modulation using transformer-based federated learning,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024, pp. 554–563.
- [2] Z. Hong, J. Wang, X. Qu, J. Liu, C. Zhao, and J. Xiao, “Federated learning with dynamic transformer for text to speech,” *arXiv preprint arXiv:2107.08795*, 2021.
- [3] G. Sun, M. Mendieta, J. Luo, S. Wu, and C. Chen, “Fedperfix: Towards partial model personalization of vision transformers in federated learning,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 4988–4998.
- [4] X. Zhang, M. Li, X. Chang, J. Chen, A. K. Roy-Chowdhury, A. T. Suresh, and S. Oymak, “Fedyolo: Augmenting federated learning with pretrained transformers,” *arXiv preprint arXiv:2307.04905*, 2023.
- [5] J. Tao, Z. Gao, and Z. Guo, “Training vision transformers in federated learning with limited edge-device resources,” *Electronics*, vol. 11, no. 17, p. 2638, 2022.
- [6] H. Li, Z. Cai, J. Wang, J. Tang, W. Ding, C.-T. Lin, and Y. Shi, “Fedtp: Federated learning by transformer personalization,” *IEEE transactions on neural networks and learning systems*, 2023.
- [7] L. Qu, Y. Zhou, P. P. Liang, Y. Xia, F. Wang, E. Adeli, L. Fei-Fei, and D. Rubin, “Rethinking architecture design for tackling data heterogeneity in federated learning,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 10 061–10 071.
- [8] S. Park, G. Kim, J. Kim, B. Kim, and J. C. Ye, “Federated split task-agnostic vision transformer for covid-19 cxr diagnosis,” *Advances in Neural Information Processing Systems*, vol. 34, pp. 24 617–24 630, 2021.
- [9] F. Almalik, N. Alkhunaizi, I. Almakky, and K. Nandakumar, “Fesvibs: Federated split learning of vision transformer with block sampling,” in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2023, pp. 350–360.
- [10] S. Baek, J. Park, P. Vepakomma, R. Raskar, M. Bennis, and S.-L. Kim, “Visual transformer meets cutmix for improved accuracy, communication efficiency, and data privacy in split learning,” *arXiv preprint arXiv:2207.00234*, 2022.
- [11] S. Oh, J. Park, S. Baek, H. Nam, P. Vepakomma, R. Raskar, M. Bennis, and S.-L. Kim, “Differentially private cutmix for split learning with vision transformer,” *arXiv preprint arXiv:2210.15986*, 2022.
- [12] D. Yao, L. Xiang, H. Xu, H. Ye, and Y. Chen, “Privacy-preserving split learning via patch shuffling over transformers,” in *2022 IEEE International Conference on Data Mining (ICDM)*. IEEE, 2022, pp. 638–647.